

Seller Behaviour & Investment: What Makes a Seller Underperform, and Where to Invest Next

A visual study of e-commerce orders, delivery and customer satisfaction.
Technical report: which seller behaviours predict excess bad reviews, and a cross-team, evidence-ranked answer to where the company should invest.

Item	Detail
Dataset	Brazilian e-commerce marketplace, 9 relational tables
Period covered	September 2016 – August 2018
Orders analysed	94,563 (after cleaning)
Sellers analysed	2,944 (1,214 with 10+ orders)
Outcome measured	Rate of 1–2 star reviews ("bad experience")
Builds directly on	the Seller Performance report – reuses its cleaning funnel and its category + delivery expected-rate model
Code files	analysis.py, make_charts.py

1. Objective

The marketplace wants to grow without breaking the customer experience. The team's shared finding so far is that delivery timing dominates satisfaction, that a small number of sellers are repeatedly worse than their category and delivery record predicts, and that ten high-revenue categories carry more than half of all bad reviews. What is still missing is the seller side of *why*: which specific things a seller does – how fast it ships, how much it sells, how wide its catalogue is, how it prices freight – actually move its bad-review rate, and how that should change where the company spends its next dollar of intervention.

This report answers the two remaining seller-side questions:

1. Which seller behaviours predict bad reviews? (Handling time, volume, catalogue breadth, freight pricing)
2. Where should the company invest to maximise growth while preserving satisfaction?

Question 1 turns the Seller Performance report's finding – that 77 sellers are repeatedly worse than expected – into something actionable: a coach cannot fix "this seller is bad," but can fix "this seller takes 3 days longer than its peers to hand a parcel to the courier." Question 2 puts every analysis's finding on one scale so leadership can rank them instead of pursuing all five at once.

DEFINITION USED THROUGHOUT (UNCHANGED FROM THE SELLER PERFORMANCE REPORT)

A **"bad experience"** is a review of 1 or 2 stars out of 5. This report reuses the Seller Performance report's cleaned order base and its category + delivery expected-rate model exactly, so that seller-level numbers in the two reports are directly comparable rather than being a second, incompatible measurement of the same sellers.

2. Data preprocessing

We rebuilt the Seller Performance report's cleaning funnel from the raw tables rather than re-using its output file, as a check that the two reports are describing the same sellers.

Cleaning step	Orders remaining
Raw orders table	99,441
Delivered only	96,478
Single-seller orders only	95,203
Has a review	94,571
Has a delivery timestamp – final analysis base	94,563

Our single-seller count (95,203) differs from the Seller Performance report's reported 95,420 by 0.2%, most likely a tie-breaking difference in how each pipeline picks the "latest" review for an order with more than one review row. It resolves before the final base: both pipelines converge on an identical **94,571** orders with a review and an identical **94,563**-order final base, and an identical **1,214**-seller ranking set (10+ orders) below – so the two reports are working from the same sellers.

2.1 Columns added for this report

Column	How it was built	Why
handling_days	order_delivered_carrier_date – order_approved_at	Time the seller itself takes to get a paid order to the courier – the seller's own speed, independent of courier transit time.
freight_ratio	total_freight ÷ total_price, per order	How expensive the seller's shipping is relative to what the customer is buying (reuses the Product & Pricing report's "freight fairness" framing).
band	delay_days grouped into the same 6 bands used in the Seller Performance report	Reused unchanged, so "expected" rates mean the same thing in both reports.
expected , excess	the Seller Performance report's category+band linear-probability model, refit on this base; excess = actual – expected	Nets out category and delivery timing so a behaviour's effect is not just re-discovering "sells a hard category" or "ships through a slow region."

handling_days could not be computed for 15 of 94,563 orders (0.02%, missing an approval or carrier-handoff timestamp); these are dropped only from the behaviour analysis, not from the funnel above.

3. Reproducing the expected-rate model

Category and delivery timing were already established as the two forces that must be controlled for before judging a seller (Seller Performance report, Sections 3–3.4) – we do not repeat that groundwork here. We refit its linear-probability model (bad-review indicator regressed on category and delivery band) on our own rebuild of the same base, as a check before using it as this report's foundation:

Check	Value
Platform bad-review rate (this base)	12.35%
Model-predicted average rate	12.35%
Difference	0.00 pp
Model R ²	0.195

The model reproduces the platform average exactly, and category+band alone explain 19.5% of the order-to-order variance in a bad review – in line with the Seller Performance report, which found delivery timing to be the single strongest driver by a wide margin. Every seller's `excess` value below is actual minus this expected rate: what remains after category and delivery are already priced in.

4. Question 1 — Which seller behaviours predict bad reviews?

For the 1,214 sellers with 10+ qualifying orders, we tested four behaviours against each seller's excess bad-review rate: average handling time, order volume, catalogue breadth (distinct categories sold), and freight ratio.

4.1 Handling time, on its own

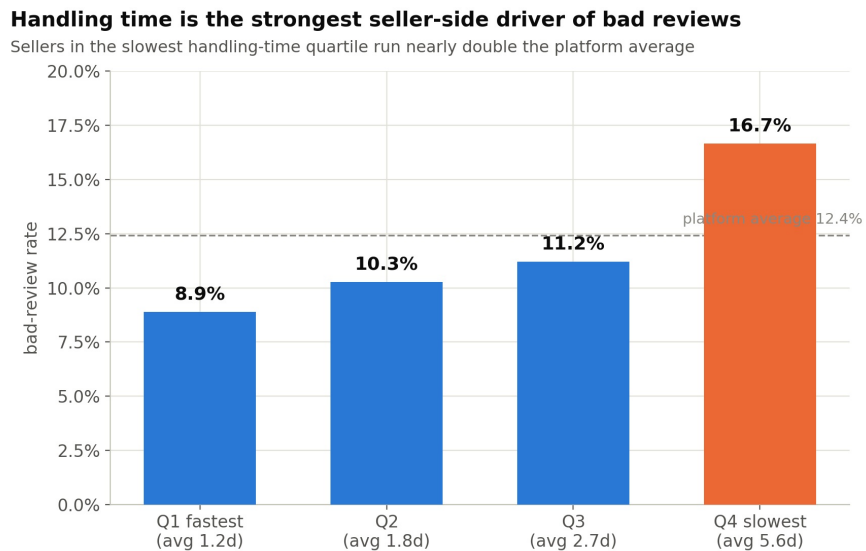


Figure 1. Actual bad-review rate by seller handling-time quartile.

Sellers in the slowest handling-time quartile (parcels sit an average of 5.6 days between payment approval and carrier handoff) run a 16.7% bad-review rate – nearly double the 8.9% of the fastest quartile (1.2 days), and well above the 12.4% platform average. The relationship is monotonic across all four quartiles.

4.2 All four behaviours, controlling for each other

Behaviour	Pearson r	p	Spearman ρ	p
Handling time	+0.244	<0.0001	+0.208	<0.0001
Catalogue breadth	+0.070	0.015	+0.092	0.001
Freight ratio	+0.040	0.168	+0.066	0.022
Volume (n_orders)	+0.018	0.537	+0.082	0.004

Bivariate correlations alone can mislead when the four behaviours move together (a high-volume seller is also more likely to sell a wide catalogue). A multiple regression puts all four in the same model against excess bad-review rate:

Behaviour	Coef.	Std. err.	t	p
Handling time (days)	+0.0079	0.0009	8.84	<0.0001
Catalogue breadth	+0.0016	0.0007	2.11	0.035
Freight ratio	+0.0151	0.0101	1.49	0.137
Volume (log orders)	+0.0007	0.0021	0.33	0.745

Model R² = 0.066; n = 1,214 sellers. Volume enters as log(n_orders) since seller size is heavily right-skewed.

Handling time dominates the other three behaviours combined

Standardized regression coefficients (category + delivery band already netted out via the excess measure)

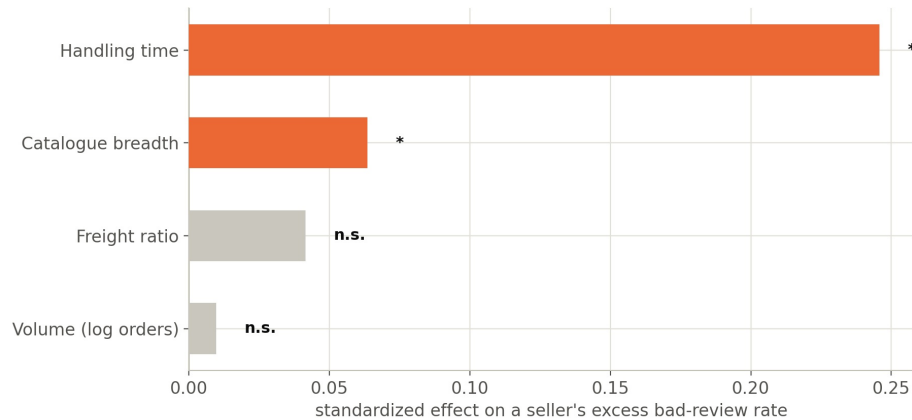


Figure 2. Standardized coefficients (comparable effect sizes) from the same regression, ranked. *** $p < 0.001$, * $p < 0.05$, n.s. = not significant.

Once the four behaviours are considered together, handling time is the only one with a large, highly significant effect – about four times the size of the next-largest. Catalogue breadth has a small but real effect: sellers with a wider catalogue run a slightly higher excess rate, consistent with more product lines meaning more ways to disappoint a customer. Freight pricing and order volume show **no reliable independent effect** once handling time and category are accounted for – their weak bivariate correlations above were mostly explained by their correlation with handling time and category, not an effect of their own.

4.3 Robustness check: does handling time survive if nothing else goes wrong?

The **excess** measure already nets out delivery band, but band is about when the parcel *arrived*; handling time is about how quickly the seller packed it. As a direct test – mirroring the same "is it just the delivery?" check used in the Seller Performance report – we split sellers into above/below-median handling time and compared their raw bad-review rate within each delivery band:

Slow-handling sellers stay worse even on parcels that arrived early or on time

The gap narrows as delay grows, and disappears once a parcel is badly late

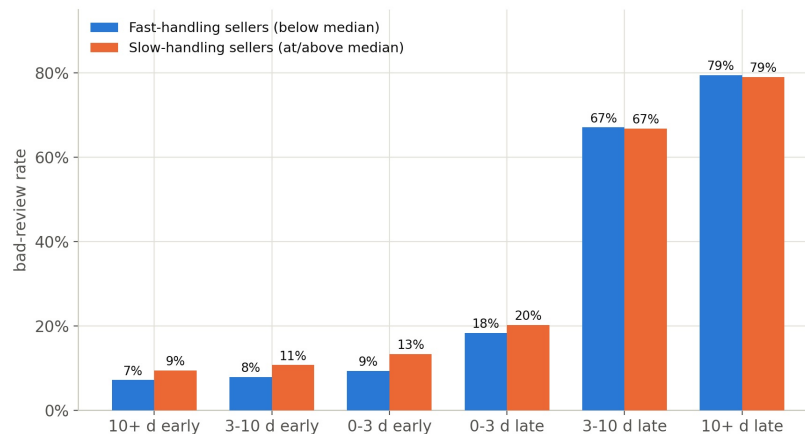


Figure 3. Bad-review rate by delivery band, split by seller handling-time group.

On parcels that arrived on time, slow-handling sellers still draw a bad review 10.0% of the time against 7.5% for fast-handling sellers – 1.33× worse, with nothing about the actual delivery to blame. The gap is widest on early and on-time parcels and steadily narrows as delay grows, disappearing entirely once a parcel is 3+ days late (67% vs. 67%, 79% vs. 79%) – at that point, exactly as the Seller Performance report found for its 77 flagged sellers, a badly late parcel angers every customer regardless of who sold it.

The slowest-handling sellers cluster among the worst offenders

Each point is one seller with 10+ qualifying orders

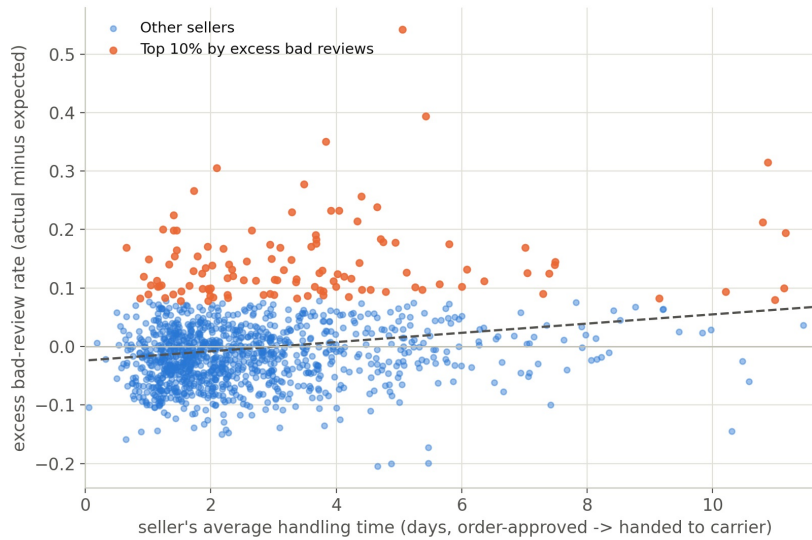


Figure 4. Every seller's handling time against its excess bad-review rate. Orange = top 10% of sellers by excess bad reviews.

The top 10% of sellers by excess bad reviews average 4.2 days of handling time against 2.7 days for everyone else ($t = 4.75$, $p < 0.0001$) – slow handling is not just statistically significant, it visibly clusters the worst offenders in Figure 4.

ANSWER TO QUESTION 1

Handling time - not volume, catalogue breadth, or freight pricing - is the seller behaviour that predicts bad reviews. It has by far the largest, most significant effect in a model that controls for the other three, it survives being tested only on parcels that arrived on time, and the gap it produces is largest exactly where a seller has no delivery excuse (early and on-time parcels) and disappears exactly where nothing a seller does matters any more (badly late parcels). Catalogue breadth has a small, genuine secondary effect. Volume and freight pricing, despite looking mildly suspicious on their own, do not survive being tested alongside the other behaviours – they are not reliable levers.

5. Question 2 — Where should the company invest?

Putting a number on each analysis's lever, in the same unit (platform bad-review-rate percentage points), is what turns five separate findings into one prioritised list. Two levers have a directly measured or reasonably estimated pp figure; two do not yet.

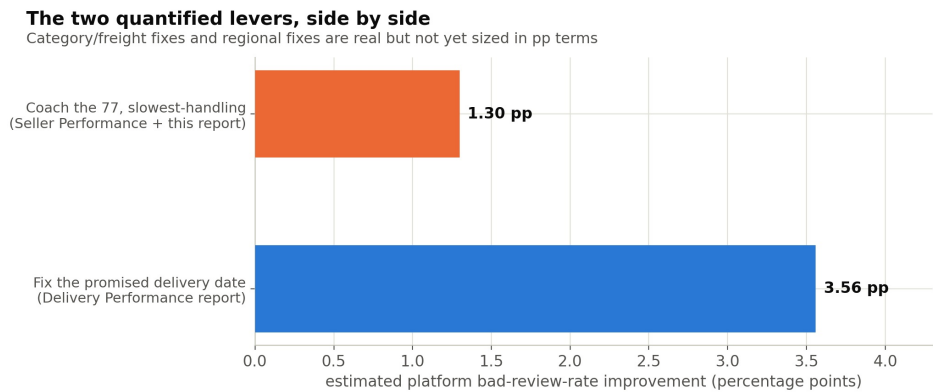


Figure 5. The two currently quantified levers, on the same scale.

Lever	Source	Est. platform pp impact	Cost	Status
Fix the promised delivery date	Delivery Performance report	~3.6 pp	None – platform-controlled	Back-of-envelope from its reported rates ¹
Coach the 77 flagged sellers (slowest-handling first)	Seller Performance report + this report	1.30 pp	None – no GMV given up	Measured (the Seller Performance report's coach-to-median model)
Category/freight-transparency fixes, 10 worst categories	Product & Pricing report	TBD	Low – listing changes	Concentration measured (52.5% revenue / 58.2% of bad reviews); pp impact not yet modelled
Regional / worst-route fixes	Regional Logistics analysis	Unknown	Unknown	Pending – questions not yet answered

1. If the 6.7% of orders that arrive late were brought to the on-time bad-review rate (9.2%) reported in the Delivery Performance report, instead of their current 62.3%: $0.067 \times (0.623 - 0.092) \approx 3.6$ pp. This is our back-of-envelope calculation from its headline numbers, not a figure quoted from that report directly – treat it as illustrative of scale, not a committed estimate.

This report's own contribution to the ladder is not a fifth, separate bar. Handling time explains a large share of why the Seller Performance report's 77 sellers are excess-flagged in the first place (mean handling time 4.2 days for the worst decile vs. 2.7 days for everyone else), so its value is in *sequencing* that coaching programme rather than adding to it: start with the slowest-handling quartile of the 77 (roughly 24% of their orders by our estimate), where a coach has a concrete, single behaviour to open the conversation with, rather than working the list in an arbitrary order.

ANSWER TO QUESTION 2

Fix delivery honesty first, then sequence seller coaching by handling time, then extend both the Product & Pricing and Regional Logistics findings to a matching pp estimate before ranking them against the first two. On the two levers the team can currently size, the delivery-promise fix (~3.6 pp) outweighs the seller-coaching programme (1.30 pp) roughly 2.7 to 1, and both cost nothing in lost revenue – they should be funded before any lever that costs GMV. Category and regional fixes are very plausibly large (the Product & Pricing report's ten worst categories already carry 58% of bad reviews) but the team cannot yet honestly rank them until they are expressed in the same pp unit as the other two.

6. What to do about it

- Fund the delivery-promise fix first.** It is the largest lever the team can currently size, and it costs nothing – the estimate is entirely within the platform's own control.
- Sequence the Seller Performance report's 77-seller coaching list by handling time, not alphabetically or by revenue.** Open every coaching conversation with the seller's own handling-time gap versus the median – it is measurable,

actionable, and (per Section 4.3) not explained away by category or delivery region.

3. **Do not add a volume cap or a catalogue-breadth policy.** Neither survives controlling for the other behaviours in Section 4.2; a policy built on either would burden sellers without moving satisfaction.
4. **Treat freight pricing as the Product & Pricing report's finding, not this report's.** Its order-level analysis of freight ratio found a real effect; this report's seller-level average freight ratio did not survive the multiple regression. The two are not contradictory – they are different units of analysis – but freight policy should be based on that order-level result, not this report's null seller-level one.
5. **Extend the Product & Pricing and Regional Logistics findings with a platform-pp estimate,** using the same "bring the worst group to the platform average" method used for the Delivery Performance and Seller Performance numbers here, so all five levers can sit on one ranked list rather than three qualitative claims and two numbers.
6. **Add handling time to the seller scorecard** alongside the Seller Performance report's already-recommended on-time bad-review rate – between the two, most of what a seller controls before and after a courier is involved is now covered.

7. Limitations

- This is an association, not a causal experiment. Handling time correlates strongly with excess bad reviews; we have not run an intervention that proves shortening it *causes* the improvement, only that it is the behaviour most worth testing first.
- The linear-probability model can, in principle, predict outside [0, 1] for extreme category/band combinations; we did not encounter this in practice but did not clip predictions, matching the approach implied in the Seller Performance report.
- Catalogue breadth is measured across a seller's full item history, not only its qualifying orders, so it reflects the seller's overall product line rather than behaviour within this specific order sample.
- Freight ratio is a seller-level average; the Product & Pricing report's order-level freight-ratio finding operates at a different grain and should not be read as contradicted by this report's seller-level null result (see Recommendation 4).
- The Q2 ladder mixes a measured figure (the Seller Performance report's 1.30 pp), a back-of-envelope estimate we calculated ourselves from the Delivery Performance report's reported rates (~3.6 pp), and two levers with no pp estimate at all. Treat the ranking's *order* as more reliable than the precise gap between these two numbers.
- This report's coaching-sequencing recommendation overlaps mechanically with the Seller Performance report's existing 77-seller list – it is a refinement of that lever, not an additive one, and should not be double-counted alongside it when totalling potential impact.
- The Regional Logistics analysis and this report's own two questions were analysed with no visibility into each other's underlying seller lists, so any overlap between "slow-handling sellers" and "sellers on the worst routes" is untested.
- Same scope as every other report on this project: one Brazilian marketplace, two years, nothing validated out of sample.

8. Reproducing this analysis

All code is included alongside this report:

`analysis.py` – rebuilds the cleaning funnel and the category+band expected-rate model from the project's cleaned parquet tables, aggregates to the seller level, and runs the correlation and regression tests in Section 4.

`make_charts.py` – produces Figures 1-5 from `analysis.py`'s output, using the same colour palette as the group's dashboard and slide deck.

Both scripts run from the project's `data/` folder (the same `orders_fact.parquet`, `items_fact.parquet` and `seller_fact.parquet` that power the group's Streamlit dashboard) plus the raw `orders_dataset.csv` and `order_items_dataset.csv` for the two timestamp fields (order approval, carrier handoff) that are not in the pre-built tables. Python, pandas, statsmodels and matplotlib were used for the same reasons given in the Seller Performance report: relational joins plus group-wise aggregation, with statsmodels added here specifically for regression p-values and standard errors.